Calibrated Perplexity for Robust AI-Generated Text Detection Across Languages and Genres

Vladimir Chaikin
MIPT
Moscow, Russia
email@phystech.edu
& *AnastasiaVoznyuk*
MIPT
Moscow, Russia
email@phystech.edu

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Abstract

This paper invistigates the problem of robust AI-generated text detection under distribution shift. While simple perplexity is a simple solution for such detection, its oppurtunities lose significantly on out-of-domain data. We propose a novel method - calibrated perplexity - which normalizes raw perplexity scores using domain-specific language model baselines. This approach constructs a stable feature space less sensitive to text length, genre, and language variations. Experiments across multiple domains and languages are going to demonstrate improved in-domain and out-of-domain detection accuracy compared to standard perplexity-based methods.

Keywords: Calibrated perplexity Cross-domain detection Robustness Machine-generated text Text classification Multilingual detection Genre adaptation Statistical methods Language modeling

1 Introduction

Fast improvement and widespread accessibility of Large Language Models (LLMs) have destroyed the lines between human-written and machine-generated text. While this technology offers benefits, it also presents significant disadvantages, including academic dishonesty, disinformation, and the fall of trust in digital content [5, 4]. This has created an inevitable need for robust methods to detect AI-generated content.

Current detectors often rely on fine-tuned classifiers or statistical signatures taken from the text [2]. Useful and computationally efficient feature is perplexity,

a measure of how "surprising" a text is to a language model. The assumption is that LLMs, which are trained to maximize likelihood, will generate text with lower perplexity compared to human writing. As it is shown in [1], a significant change comes from distribution shift. The perplexity of a text is not an absolute value; it is highly dependent on the domain (e.g., news, scientific articles, social media) and language it was trained on. A model fine-tuned for one genre often fails on another, causing unreliable detection in real world, where the target domain is unknown.

Recent works have attempted to solve this by developing zero-shot detectors that are more robust to unseen prompts or models. For instance, Binoculars [4] uses a ratio of scores from two different LLMs for calibration. Similarly, research on multilingual and cross-domain detection highlights the necessity of adapting methods to diverse data distributions [3]. Despite these advances, a method for calibrating perplexity for each domain itself to make it robust across different genres and languages remains still not-explored.

In this paper, we face the challenge of domain shift and try to avoid its effect on detection. We propose a novel method called calibrated perplexity, which normalizes perplexity scores against domain-specific baselines. The final goal is not to find out one universal calibrated perplexity score, but to get a set of parameters to have an opportunity to learn fastly new zero-shot models. Our approach includeses training KenLM n-gram models on text from multiple domains and using these models to compute a relative, rather than absolute, measure of text predictability. The core novelty of our work lies in transforming a domain-dependent score into a stable feature for classification. To check our method, we conduct experiments across multiple domains and languages, comparing in-domain and out-of-domain detection.

2 Problem statement

Let $\mathbf{W}$ be our alphabet, tokens consist of characters from that alphabet.

Let

$$\mathbb{D} = \left\{ [t_j]_{j=1}^{n} \;\middle|\; t_j \in \mathbf{W},\ n \in \mathbb{N} \right\}$$

be the space of our documents.

We have a set of N documents

$$\mathbf{D} = \bigcup_{i=1}^{N} D^i, \quad D^i \in \mathbb{D}.$$

We assume that each document may contain fragments generated by one of K text-generative models, or be fully written by a human. Accordingly, we introduce $K+1$ labels for multiclass classification, where 0 denotes a human-written fragment and $\{1, \ldots, K\}$ denote fragments generated by the corresponding language model. Let $\mathbf{C}$ be the set of these labels.

To capture the statistical properties of text fragments, we consider a probabilistic model of the language. For a given fragment $F = [t_a, t_{a+1}, \ldots, t_b]$ we

define its probability as the product of conditional probabilities of each token given its $n-1$ predecessors:

$$P(F) = \prod_{i=a}^{b} P(t_i \mid t_{i-1}, \ldots, t_{i-n+1}).$$

The choice of such an n-gram model is motivated by Hypothesis 1 in [3], which states that generative language models tend to sample from the head of the token distribution, i.e., they prefer high-probability continuations. This makes n-gram statistics a natural baseline for distinguishing human-written from machine-generated text.

The log-probability of a fragment is then

$$\log P(F) = \sum_{i=a}^{b} \log P(t_i \mid t_{i-1}, \ldots, t_{i-n+1}),$$

and the perplexity is defined as the inverse geometric mean of the token probabilities:

$$\mathrm{PPL}(F) = P(F)^{-\frac{1}{|F|}} = 2^{-\frac{1}{|F|}\log_2 P(F)},$$

where $|F| = b - a + 1$ is the length of the fragment in tokens.

Perplexity quantifies how "surprising" a fragment is under the language model: lower values indicate that the fragment is more predictable.

Given a training set of documents $\mathbf{D}_{\text{train}} \subset \mathbb{D}$ with known labels, we learn the conditional probabilities $P(t_i \mid t_{i-1}, \ldots, t_{i-n+1})$ by maximizing the likelihood of the training data. This leads to the following optimization problem:

$$\mathcal{L}(\boldsymbol{\theta}) = -\sum_{F \in \mathbf{D}_{\text{train}}} \log P(F) \to \min_{\boldsymbol{\theta}},$$

where $\boldsymbol{\theta}$ denotes the parameters of the n-gram model (the conditional probability tables).

The training objective is the negative log-likelihood of the observed documents, which is equivalent to minimizing the cross-entropy between the model distribution and the empirical distribution of the training data. This optimization is typically performed using count-based estimation with appropriate smoothing to handle sparse n-gram occurrences.

For classification, we introduce a threshold τ_d for each domain d. A fragment F is classified as machine-generated if:

$$\mathrm{PPL}_{\mathcal{M}_d}(F) < \tau_d,$$

and as human-written otherwise. The threshold τ_d is selected on a validation set to maximize the F1-score, balancing precision and recall.

3 Method

3.1 Overview

We propose an ensemble-based method for AI-generated text detection that operates without knowledge of the test text's domain. The method consists of three stages:

1. **Training:** Train separate n-gram language models on different domains using only human-written texts.
2. **Threshold calibration:** For each domain model, select a classification threshold that maximizes F1-score on a validation set from the same domain.
3. **Ensemble inference:** For a new text, compute perplexities from all domain models and combine their predictions using weighted voting.

3.2 Notation

Let $\mathcal{D} = \{d_1, d_2, \ldots, d_M\}$ be a set of M domains (e.g., news, essays, scientific texts). For each domain d, we have:

- Training set $\mathbf{T}_d \subset \mathbb{D}$ consisting of human-written texts from domain d
- Validation set $\mathbf{V}_d$ with balanced human and machine texts from domain d
- Test set $\mathbf{X}_d$ held-out from the same distribution as validation

For a text x, let $\mathrm{PPL}_{\mathcal{M}_d}(x)$ denote the perplexity computed by the language model trained on domain d.

3.3 Model Training

For each domain $d \in \mathcal{D}$, we train a trigram language model $\mathcal{M}_d$ using KenLM with modified Kneser-Ney smoothing. The training data consists exclusively of human-written texts from that domain:

$$\mathcal{M}_d = \mathrm{KenLM}(\mathbf{T}_d), \quad \text{where } \mathbf{T}_d = \{x \in \mathbb{D} \mid \mathrm{source}(x) = \mathrm{human}, \mathrm{domain}(x) = d\}.$$

The training objective is to minimize the negative log-likelihood of the training corpus:

$$\mathcal{L}(\mathcal{M}_d) = -\sum_{x \in \mathbf{T}_d} \log P_{\mathcal{M}_d}(x) \to \min.$$

3.4 Threshold Calibration

For each model $\mathcal{M}_d$, we find an optimal threshold τ_d that maximizes the F1-score on the validation set $\mathbf{V}_d$:

$$\tau_d = \arg\max_{\tau} \mathrm{F1}\left(\mathbf{1}[\mathrm{PPL}_{\mathcal{M}_d}(x) < \tau],\ y(x)\right),$$

where $y(x) = 1$ for machine-generated texts and $y(x) = 0$ for human-written texts.

The F1-score is defined as the harmonic mean of precision and recall:

$$\mathrm{F1} = 2 \cdot \frac{\mathrm{precision} \cdot \mathrm{recall}}{\mathrm{precision} + \mathrm{recall}}.$$

In practice, we search over 200 evenly spaced thresholds between the minimum and maximum perplexity values observed on the validation set.

3.5 Ensemble Inference

For a test text x with unknown domain, we compute perplexities $p_d = \mathrm{PPL}_{\mathcal{M}_d}(x)$ for all $d \in \mathcal{D}$. These perplexities are then used in two complementary ensemble strategies.

3.5.1 Strategy A: Machine-Oriented Ensemble (Unweighted Voting)

This ensemble focuses on detecting machine-generated texts. Each model makes an individual prediction $h_d^{\mathrm{machine}}(x) = \mathbf{1}[p_d < \tau_d]$, where 1 indicates "machine-generated". The final decision is based on simple majority voting:

$$H_{\mathrm{machine}}(x) = \mathbf{1}\left[\sum_{d \in \mathcal{D}} h_d^{\mathrm{machine}}(x) \geq \lceil M/2 \rceil\right],$$

where $M = |\mathcal{D}|$ is the number of domains. This strategy is effective when the input text is likely to be machine-generated, as it requires a simple majority of models to agree.

3.5.2 Strategy B: Human-Oriented Ensemble (Perplexity-Weighted Voting)

This ensemble focuses on detecting human-written texts. The key insight is that a text originating from a particular domain will yield minimal perplexity for the model trained on that domain. Therefore, we sort the domains by their perplexity in ascending order and assign decreasing weights:

$$w_{(1)} \geq w_{(2)} \geq \cdots \geq w_{(M)}, \quad \sum_{k=1}^{M} w_k = 1,$$

where (k) denotes the index of the model with the k-th smallest perplexity. In our implementation, we use:

$$[w_1, w_2, w_3, w_4, w_5, w_6, w_7] = [0.50, 0.20, 0.10, 0.07, 0.05, 0.04, 0.04].$$

Each model makes an individual prediction $h_d^{\text{human}}(x) = \mathbf{1}[p_d \geq \tau_d]$, where 1 indicates "human-written". The weighted sum is:

$$S_{\text{human}}(x) = \sum_{k=1}^{M} w_k \cdot h_{(k)}^{\text{human}}(x),$$

and the final decision is:

$$H_{\text{human}}(x) = \mathbf{1}\left[S_{\text{human}}(x) \geq \theta\right],$$

where θ is a global threshold (typically $\theta = 0.5$). This strategy prioritizes models that best fit the input text (lowest perplexity), making it particularly sensitive to texts that are likely to be human-written.

3.5.3 Combined Approach

The two strategies complement each other:

- Strategy A (unweighted majority) excels at catching machine-generated texts, as any sign of machine generation from multiple models is sufficient.
- Strategy B (perplexity-weighted) excels at confirming human-written texts, as a human text from a known domain will be recognized by its matching model.

In practice, we can either:

1. Apply both strategies and combine their outputs (e.g., by averaging),
2. Select the appropriate strategy based on the application requirements (e.g., use Strategy A for spam filtering, Strategy B for academic integrity checks),
3. Use the weighted sum $S_{\text{human}}(x)$ directly as a confidence score.

The optimal threshold θ for Strategy B is selected on the validation set by maximizing the balanced accuracy:

$$\theta^* = \arg\max_{\theta} \text{BalancedAcc}\left(\mathbf{1}[S_{\text{human}}(x) \geq \theta],\ y(x)\right),$$

where $y(x) = 1$ for human-written texts.

3.6 Properties

The proposed method has the following properties:

- **Domain-agnostic inference:** No prior knowledge of the test text's domain is required.
- **Interpretability:** Each model provides a clear per-domain prediction; low perplexity indicates that the text "belongs" to that domain.
- **Robustness:** Weighted voting reduces the impact of models that perform poorly on the input text.
- **Scalability:** Adding a new domain requires training one additional KenLM model and computing its validation accuracy.

3.7 Optimization Problem

$$\sum_{i=1}^{N_d} \sum_{x_j \in X_i} (M_{\mu_i,i}(x) - y_{ij})^2 + \sum_{i<j\leq N_d} (\mathrm{TH}_{M_{\mu_i,i}}(x) - \mathrm{TH}_{M_{\mu_j,j}}(x))^2 \to \min_{\bar{\mu}}$$

We minimize the variance of thresholds in ensemble across different domains for the same text x, making the score domain-invariant

Our ensemble approach approximates this ideal by weighting models according to their reliability. When all models give similar perplexities, the confidence is high; when they disagree, the weighted sum provides a robust decision.

4 Experiments

4.1 Experimental Setup

We conduct experiments on a collection of Russian-language texts spanning seven domains: news, scientific texts, stories, articles, and 3 "hard" domains (hard_news, hard_science, hard_essays). The dataset is balanced with approximately 3,500 human-written and 3,500 machine-generated texts per domain. Texts shorter than 50 words are filtered out. All models are trained using KenLM with trigram order ($n = 3$) and modified Kneser-Ney smoothing.

4.2 Experiment 1: Single-Domain Training and In-Domain Testing

For each domain, we train a KenLM model exclusively on human-written texts from that domain. We then evaluate the model on a held-out test set from the same domain, using a threshold optimized by F1-score on a validation set.

Results: In-domain performance varies significantly across domains, with F1-scores ranging from 0.85 (AINL science) to 0.53 (reviews). The highest balanced accuracy (0.85) is achieved on scientific texts, while reviews prove the most challenging due to short and stylistically homogeneous texts.

4.3 Experiment 2: Cross-Domain Testing

We evaluate each model on test sets from all other domains, using thresholds calibrated on the source domain.

Results: Cross-domain performance drops substantially. For example, the model trained on news achieves an F1-score of only 0.69 when tested on scientific texts, compared to 0.85 in-domain. This confirms that perplexity thresholds are domain-dependent and cannot be directly transferred.

4.4 Experiment 3: Generalization on Mixed-Domain Data

We combine all domains into a single balanced dataset (7,056 texts total, split 70/15/15 for train/validation/test). A single KenLM model is trained on the combined training set.

Results: The unified model achieves a balanced accuracy of 0.68 and F1-score of 0.68. While this outperforms cross-domain transfer, it still lags behind in-domain performance on individual domains, indicating that domain heterogeneity remains a challenge.

4.5 Experiment 4: Tokenization and Pruning Optimization

We experiment with SentencePiece tokenization (vocab_size=8000) and n-gram pruning ($-prune$ 2 2 2) to improve model stability across domains.

Results: Combined optimization yields a modest improvement of 1-2% in balanced accuracy (from 0.68 to 0.69), suggesting that while helpful, tokenization and pruning alone are insufficient to fully resolve domain shift.

4.6 Experiment 5: Ensemble Methods

We implement four ensemble strategies combining the seven domain-specific models:

Table 1: Comparison of ensemble methods on the mixed-domain test set.

Method	**Acc**	**F1**	**Balanced Acc**	**Human/AI**
Simple majority voting	0.63	0.65	0.61	32% / 91%
Weighted voting (by val accuracy)	0.68	0.65	0.68	78% / 58%
Weighted voting (by balanced acc)	0.68	0.68	0.69	73% / 66%
Perplexity-weighted voting	0.63	0.47	0.63	93% / 33%

Observations:

- Simple majority voting strongly favors machine detection (91% AI recall) but fails on human texts (32%).
- Weighted voting by balanced accuracy achieves the most balanced performance, with human and AI detection rates of 73% and 66%, respectively.
- Perplexity-weighted voting overfits to human texts, achieving 93% human recall at the cost of only 33% AI recall.

4.7 Summary of Key Findings

1. **Domain shift is significant:** Optimal thresholds vary by a factor of 2-3 across domains (from 8.28 to 12.10).
2. **Single-domain models do not generalize:** Cross-domain F1-scores drop by 0.15–0.30 compared to in-domain.
3. **Tokenization and pruning provide marginal gains:** Only 1–2% improvement on mixed-domain data.
4. **Weighted ensemble with balanced accuracy is most robust:** Achieves the highest balanced accuracy (0.69) and most balanced human/AI detection rates (73% / 66%).

5 Discussion

TODO

6 Conclusion

TODO

References

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A Appendix / supplemental material

A.1 Additional experiments / Proofs of Theorems

TODO